Worksheet #18

Week 9, Day 2: Support vector machines; kernels

Name: _____
NetID: _____
Date: _____

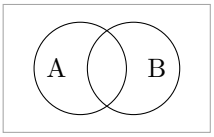
Guided checkpoints

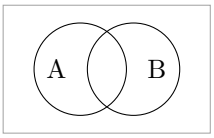
For full credit, fill in the checkpoints. If you skip a checkpoint, you must include your own notes for full credit.

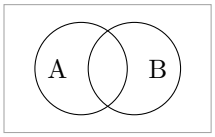
1. Jaccard and SMC, in Pictures

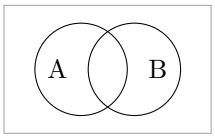
Shade the numerator's region in the top box and the denominator's region in the bottom box.

$$\text{Jaccard} = \frac{\quad}{\quad} \qquad \text{SMC} = \frac{\quad}{\quad}$$









Which region does SMC count that Jaccard ignores?

2. Teach the Tree New Words

Open the movie-reviews demo in the Python Yard (`movie-reviews.ipynb`). Add three words to the word lists, re-run, and record what changed.

	Word added	Which list (neg. / meh / pos.)	New test accuracy
Start	*(none yet)*		*51%*

New root question: _____

3. Is 51% Good?

There are three labels, and each has 1,500 reviews. What accuracy would you expect from guessing a label at random? With that in mind, is 51% good?

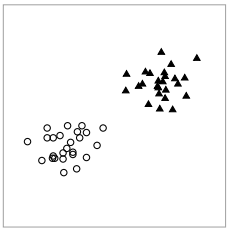
Where did each label come from? That is, what is the provenance of these labels? Could they be incorrect?

4. SVM and SVC

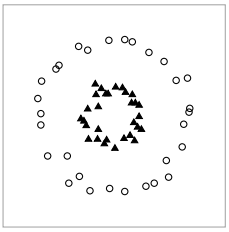
What does SVM stand for?

5. Linear or Kernel?

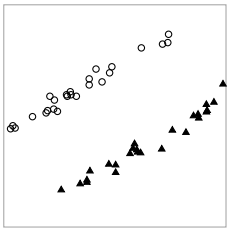
Each panel shows two classes, ▲ and ○. Circle **L** if one straight line can separate them (a linear SVM is enough), or **K** if the SVM needs a kernel.



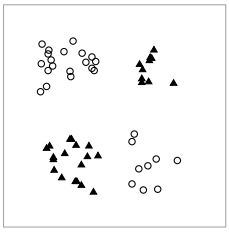
L / K



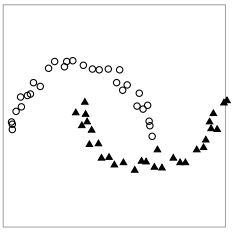
L / K



L / K



L / K



L / K

6. Strengths and Weaknesses

Classifier	One strength	One weakness
Decision tree		
k-NN		
Linear SVM		